

Qatar-2b

R-1845

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_150401 · created 2026-08-03T07:32:06+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2457113.815179 +/- 7.8e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.137 +/- 0.029
Transit depth (Rp/Rs)^2	1.87 +/- 0.78 [%]
Orbital Inclination (inc)	85.47 +/- 1.7
Airmass coefficient 1 (a1)	0.962 +/- 0.027
Airmass coefficient 2 (a2)	0.025 +/- 0.021
Scatter in the residuals of the lightcurve fit is	2.77 %
Best Comparison Star	#2 - [28, 281]
Optimal Aperture	3.08
Optimal Annulus	12.68
Transit Duration (day)	0.0659 +/- 0.0076

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.137 $\pm$ 0.029 vs 0.16327 (deviation 0.91 σ)
Duration fit vs published	0.0659 d vs 0.0767 d (deviation 1.42 σ)
Mid-time O-C	0.3 min
Depth SNR	4.7
Residual scatter	2.77 %

Light curve

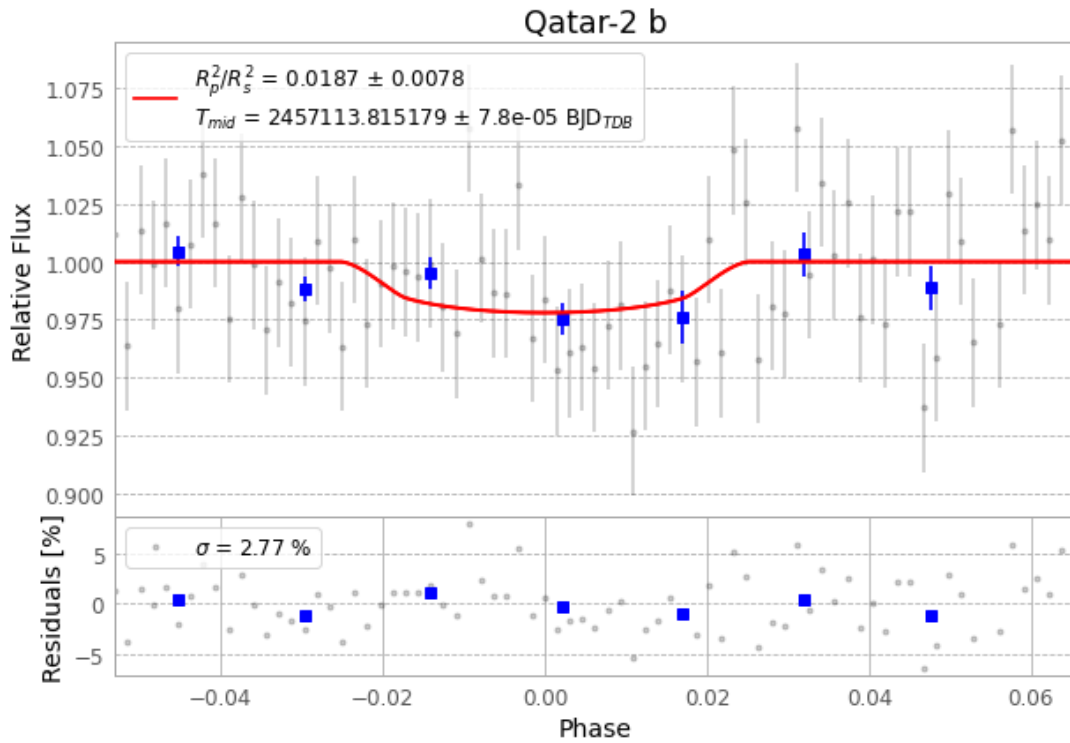


Figure 1 — FinalLightCurve_Qatar-2 b_2015-03-31.png (SHA-256 a2622450585ed7e7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [43, 214], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ee8780f0968ac88f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), Qatar-2150401054212.FITS → Qatar-2150401093012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-150401.log (69143246bda2...), FinalLightCurve_Qatar-2 b_2015-03-31.png (a2622450585e...), FinalLightCurve_Qatar-2 b_2015-03-31.csv (5b969d3bc161...)

Compute resources

Wall time 145.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 07:32Z.